

ENUGALA VAMSHIDHAR REDDY

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Profile

Mechanical Engineer specialising in structural dynamics and NVH, with experience in FEA, vibration testing, and simulation-test correlation on electric powertrain components. Built and validated FE models in ANSYS Mechanical for modal and harmonic response analysis, applied FFT and FRF-based signal processing to identify dominant excitation sources, and developed neural surrogate models (TensorFlow, Keras) replacing computationally expensive NASTRAN simulations.

Skills

Tools: ANSYS Mechanical, PyMechanical, NASTRAN, Hypermesh, Simcenter 3D, SolidWorks, Python, IronPython, NX Open API, MATLAB, TensorFlow

Domains: Structural Analysis (Static, Dynamic & Thermo-elastic Loadcases), Finite Element Analysis (FEA), Simulation-Test Correlation, NVH & Vibration Analysis, Metal & Composite Structures, Vibroacoustics, Surrogate Modeling, Neural Networks, Bayesian Optimization, DIMGP

Work Experience

Internship

04/2026 - Present

Mercedes-Benz AG, Untertürkheim, Stuttgart, Germany

- Preprocessed a V8 engine model in Hypermesh to set up DOE-based NASTRAN simulations, generating structural FRF training data across varying design parameters.
- Automated NASTRAN job execution and PCH/FIF output parsing in Python, streamlining structural FRF data extraction and preprocessing pipelines.
- Built and trained TensorFlow/Keras neural surrogate models predicting structural FRFs across hundreds of frequency points, benchmarked against DIMGP models.

Internship

09/2025 - 03/2026

Valeo Schalter und Sensoren GmbH, Bietigheim-Bissingen, Germany

- Automated vibroacoustic simulation model setup in Simcenter 3D using Python and NX Open, eliminating repetitive manual configuration steps.
- Configured sound sources, microphone arrays, and acoustic environments for airborne noise simulations, supporting team noise prediction studies.
- Built Python tools for SPL and acoustic energy visualization, adopted by the wider simulation team for noise prediction reporting.

Master Thesis, Simulation Department

03/2025 - 08/2025

Valeo eAutomotive Germany GmbH, Erlangen, Germany

- Built and validated an electric motor FE model in ANSYS Mechanical for modal and harmonic response analysis, delivering the core structural model for NVH evaluation.
- Derived electromagnetic excitation forces from EM simulations and experimental measurements to characterize motor vibration sources.
- Correlated simulation and physical test results from accelerometer measurements, improving NVH prediction reliability.

Working Student, Simulation Department

08/2024 - 02/2025

Valeo eAutomotive Germany GmbH, Erlangen, Germany

- Built automated pre- and post-processing workflows in ANSYS Mechanical using Python and IronPython, deployed across the simulation team.
- Ran thermal simulations of electric modules and correlated results against experimental warpage measurements to validate thermo-elastic model accuracy.
- Refined non-linear material model parameters iteratively based on test correlation, improving simulated-to-measured agreement.

Selected Projects & Hackathons

- **Agentic FEA | 3D Aerospace Bracket Modal Analysis (Ansys 2026 R1 & PyMechanical):** Automated end-to-end vibration pipeline via Claude MCP; scripted CAD setup, solid meshing, bolted constraints, and Block Lanczos eigensolving.
- **Hack A-Bot Hackathon 2026 Winner (Siemens Healthineers):** Engineered an interactive AI system for surgical robotic apparatus in operation theaters, eliminating touch contamination.
- **Topology Optimization & Motorsports (FAU):** Designed lightweight chassis structures and performed FEA topology optimization under stiffness constraints.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

2022 - 2026

M.Sc. Computational Engineering (GPA: 2.0), Focus: Structural Mechanics & Dynamics, Machine Learning

Kakatiya Institute of Technology and Science, India

2018 - 2022

B.Tech. Mechanical Engineering (German GPA: 1.7)

Work Authorization & Availability: Unrestricted German work authorization (EU Blue Card eligible - §20/§18g AufenthG) Immediate availability

Languages: English (Fluent / C2), German (Intermediate / B1, advancing), Telugu (Native) **Other:** Driving License Class B